

Problem

Rework Prob. 14.25 if the elements are connected in parallel.

Step-by-step solution

Step 1 of 8

Refer to problem 14.25P in the textbook.

Determine the admittance of the parallel RLC circuit.

$$Y = R + j\omega C + \frac{1}{j\omega L}$$

$$Y = R + j\left(\omega C - \frac{1}{\omega L}\right) \dots (1)$$

$$Y = R + j\left(\frac{\omega^2 LC - 1}{\omega L}\right)$$

$$Y = \frac{\omega LR + j(\omega^2 LC - 1)}{\omega L} \dots (2)$$

Resonance will occur when imaginary part of Y in equation (1) is zero.

$$\omega C - \frac{1}{\omega L} = 0$$

$$\omega = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

Determine the resonance frequency ω_o .

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Substitute $L = 40 \text{ mH}$, and $C = 1 \mu\text{F}$.

$$\begin{aligned} \omega_o &= \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} \\ &= 5 \text{ krad/s} \end{aligned}$$

Step 2 of 8

Write the expression for the impedance the parallel RLC circuit using equation (2).

$$Z = \frac{1}{Y}$$

$$Z = \frac{1}{\left(\frac{\omega LR + j(\omega^2 LC - 1)}{\omega L}\right)}$$

$$Z = \frac{\omega L}{\omega LR + j(\omega^2 LC - 1)} \dots (3)$$

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Determine the impedance at resonance frequency ($\omega = \omega_o$) using equation (3).

$$\begin{aligned} Z(\omega_o) &= \frac{\omega_o L}{\omega_o LR + j(\omega_o^2 LC - 1)} \\ &= \frac{\omega_o L}{\omega_o LR + j0} \\ &= \frac{1}{R} \end{aligned}$$

Substitute $R = 2 \text{ k}\Omega$.

$$\begin{aligned} Z(\omega_o) &= \frac{1}{2 \times 10^3} \\ &= 0.5 \text{ m}\Omega \end{aligned}$$

Hence, the impedance at resonance frequency is, $0.5 \text{ m}\Omega$.

Step 4 of 8

Determine the admittance at one-fourth of the resonance frequency ($\omega = \frac{\omega_o}{4}$) using equation (1).

$$Y\left(\frac{\omega_o}{4}\right) = R + j\left(\left(\frac{\omega_o}{4}\right)C - \frac{1}{\frac{\omega_o}{4}L}\right)$$

Substitute $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $\omega_o = 5 \text{ krad/s}$.

$$\begin{aligned} Y\left(\frac{\omega_o}{4}\right) &= (2 \times 10^3) + j\left(\left(\frac{5 \times 10^3}{4}\right)(1 \times 10^{-6}) - \frac{1}{\left(\frac{5 \times 10^3}{4}\right)(40 \times 10^{-3})}\right) \\ &= (2 \times 10^3) + j(1.25 \times 10^{-3} - 0.02) \\ &= 2,000 - j0.01875 \end{aligned}$$

Determine the impedance at one-fourth of the resonance frequency $Z\left(\frac{\omega_o}{4}\right)$.

$$\begin{aligned} Z &= \frac{1}{Y} \\ Z\left(\frac{\omega_o}{4}\right) &= \left[\frac{1}{(2,000 - j0.01875)} \times (2,000 + j0.01875)\right] \\ Z\left(\frac{\omega_o}{4}\right) &= \left[\frac{2,000 + j0.01875}{4 \times 10^6}\right] \\ Z\left(\frac{\omega_o}{4}\right) &= (500 \times 10^{-6} + j0.00468 \times 10^{-6}) \Omega \\ Z\left(\frac{\omega_o}{4}\right) &= (500 + j0.00468) \mu\Omega \end{aligned}$$

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Hence, the impedance at one-fourth of the resonance frequency $Z\left(\frac{\omega_o}{4}\right)$ is,

$$\boxed{(500 + j0.00468) \mu\Omega}.$$

Step 6 of 8

Determine the impedance at one-half of the resonant frequency $Z\left(\frac{\omega_o}{2}\right)$ using equation (3).

$$Z\left(\frac{\omega_o}{2}\right) = \frac{\left(\frac{\omega_o}{2}\right)L}{\left(\frac{\omega_o}{2}\right)LR + j\left(\left(\frac{\omega_o}{2}\right)^2 LC - 1\right)}.$$

Substitute $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $\omega_o = 5 \text{ krad/s}$.

$$\begin{aligned} Z\left(\frac{\omega_o}{2}\right) &= \frac{\left(\frac{5 \times 10^3}{2}\right)(40 \times 10^{-3})}{\left(\frac{5 \times 10^3}{2}\right)(40 \times 10^{-3})(2 \times 10^3) + j\left(\left(\frac{5 \times 10^3}{2}\right)^2 (40 \times 10^{-3})(1 \times 10^{-6}) - 1\right)} \\ &= \frac{100}{(200 \times 10^3) + j(0.250 - 1)} \\ &= \left(\frac{100}{200 \times 10^3 - j0.75}\right) \times \left(\frac{200 \times 10^3 + j0.75}{200 \times 10^3 + j0.75}\right) \\ &= \left(\frac{100}{4 \times 10^{10}}\right) \times (200 \times 10^3 + j0.75) \\ &= (25 \times 10^{-10})(200 \times 10^3 + j0.75) \\ &= 500 \times 10^{-6} + j0.00187 \times 10^{-6} \\ Z\left(\frac{\omega_o}{2}\right) &= (500 + j0.00187) \text{ }\mu\Omega \end{aligned}$$

Therefore, the impedance at one-half of the resonant frequency $Z\left(\frac{\omega_o}{2}\right)$, is

$$\boxed{(500 + j0.00187) \text{ }\mu\Omega}.$$

Step 7 of 8

Determine the impedance at twice of the resonant frequency $Z(2\omega_o)$, using equation (3).

$$Z(2\omega_o) = \frac{(2\omega_o)L}{(2\omega_o)LR + j((2\omega_o)^2 LC - 1)}.$$

Substitute $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $\omega_o = 5 \text{ krad/s}$.

$$\begin{aligned} Z(2\omega_o) &= \frac{(2 \times 5 \times 10^3)(40 \times 10^{-3})}{(2 \times 5 \times 10^3)(40 \times 10^{-3})(2 \times 10^3) + j((2 \times 5 \times 10^3)^2 (40 \times 10^{-3})(1 \times 10^{-6}) - 1)} \\ &= \frac{400}{(800 \times 10^3) + j(4 - 1)} \\ &= \frac{400}{(800 \times 10^3) + j3} \times \left(\frac{(800 \times 10^3) - j3}{(800 \times 10^3) - j3}\right) \\ &= \frac{400((800 \times 10^3) - j3)}{(64 \times 10^{10})} \\ &= (6.25 \times 10^{-10})((800 \times 10^3) - j3) \\ &= 5000 \times 10^{-7} - j18.75 \times 10^{-10} \\ Z(2\omega_o) &= 500 - j0.00187 \text{ }\mu\Omega \end{aligned}$$

Therefore, the impedance at twice of the resonant frequency $Z(2\omega_o)$ is,

$$\boxed{(500 - j0.00187) \text{ }\mu\Omega}.$$

Step 8 of 8

Determine the impedance at four times of the resonant frequency $Z(4\omega_o)$, using equation (3).

$$Z(4\omega_o) = \frac{(4\omega_o)L}{(4\omega_o)LR + j((4\omega_o)^2 LC - 1)}.$$

Substitute $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $\omega_o = 5 \text{ krad/s}$.

$$\begin{aligned} Z(4\omega_o) &= \frac{(4 \times 5 \times 10^3)(40 \times 10^{-3})}{(4 \times 5 \times 10^3)(40 \times 10^{-3})(2 \times 10^3) + j((4 \times 5 \times 10^3)^2(40 \times 10^{-3})(1 \times 10^{-6}) - 1)} \\ &= \frac{800}{(1600 \times 10^3) + j(16 - 1)} \\ &= \frac{800}{(1600 \times 10^3) + j15} \times \left(\frac{(1600 \times 10^3) - j15}{(1600 \times 10^3) - j15} \right) \\ &= \frac{800((1600 \times 10^3) - j15)}{(256 \times 10^{10})} \\ &= 3.125 \times 10^{-10} ((1600 \times 10^3) - j15) \\ &= (500 \times 10^{-6} - j0.04687 \times 10^{-6}) \Omega \end{aligned}$$

$$Z(4\omega_o) = (500 - j0.004687) \mu\Omega$$

Therefore, the impedance at four times of the resonant frequency $Z(4\omega_o)$, is,

$$\boxed{(500 - j0.004687) \mu\Omega}.$$